Class 6 Science – Chapter 7: Temperature and Its Measurement

नीचे आपके अपलोड किए गए अध्याय के पृष्ठ 90–97 के चित्रों, उदाहरणों, activities और सभी दिखाई देने वाले प्रश्नों के आधार पर उत्तर simple English में दिए गए हैं। इसके बाद प्रत्येक topic से अतिरिक्त महत्वपूर्ण प्रश्न-उत्तर और numerical questions भी दिए गए हैं।

07-temperature-and-its-measurement.pdf

1. Hot and Cold (Page 90)

The pictures show a glass of cold water, a pot being heated, an ice cream, a burning candle, a bowl of hot food and a glass of iced drink.

Q1. Name the items in the given pictures and classify them as hot or cold.

Answer:

Picture	Item	Classification
(a)	Cold water	Cold
(b)	A pot being heated	Hot
(c)	Ice cream	Cold
(d)	Burning candle	Hot
(e)	Hot food	Hot
(f)	Iced drink	Cold

Q2. How do you find out the hotness or coldness of an object?

Answer: We may feel whether an object is hot or cold, but we should use a thermometer to measure its temperature accurately. We should never touch very hot objects to test them.

Q3. List things from your surroundings that are hot and cold.

Answer:

- Hot things: Hot tea, hot food, a heated pan and warm water.
- Cold things: Ice, ice cream, cold water and chilled juice.

2. Temperature (Page 91)

Q1. What is temperature?

Answer: Temperature is the measure of the degree of hotness or coldness of a body.

Q2. How can we compare the temperature of two bodies?

Answer: We can measure their temperatures using a thermometer and compare the readings.

Q3. Can our sense of touch always tell us the correct temperature?

Answer: No. Our sense of touch may sometimes give a wrong impression. A thermometer gives a more accurate measurement.

Activity: Three Containers of Water

The activity uses ice-cold water, normal tap water and warm water in three containers.

Q4. What happens when both hands are placed in normal water after one hand has been kept in cold water and the other in warm water?

Answer: The hand that was in cold water feels the normal water warm, while the hand that was in warm water feels the normal water cold.

Q5. What does this activity show?

Answer: It shows that our sense of touch is not always reliable for judging temperature.

Q6. Why do we need thermometers?

Answer: We need thermometers to measure temperature accurately and safely.

HOTS: Think and Answer

Q7. Do you feel more active in summer or winter?

Answer: This depends on the person. I may feel more active in winter because the weather is cooler and more comfortable.

3. Temperature Scales (Pages 91–92)

There are three commonly used temperature scales in the chapter: Celsius, Fahrenheit and Kelvin.

A. Celsius Scale

Q1. What is the Celsius scale?

Answer: The Celsius scale is a temperature scale in which water freezes at 0°C and boils at 100°C under standard atmospheric pressure.

Q2. What are the lower and upper fixed points on the Celsius scale?

Answer:

- Lower fixed point: 0°C
- Upper fixed point: 100°C

Q3. Into how many equal divisions is the interval between these points divided?

Answer: It is divided into 100 equal divisions.

Q4. Who is associated with the Celsius scale?

Answer: Anders Celsius, a Swedish astronomer, physicist and mathematician, developed the Celsius temperature scale.

B. Fahrenheit Scale

Q1. What is the Fahrenheit scale?

Answer: The Fahrenheit scale is a temperature scale used in some countries, including the United States.

Q2. What are the freezing and boiling points of water on the Fahrenheit scale?

Answer:

- Freezing point: 32°F
- Boiling point: 212°F

Q3. Into how many equal divisions is the interval between these points divided?

Answer: It is divided into 180 equal divisions.

Q4. Who proposed the Fahrenheit scale?

Answer: Daniel Gabriel Fahrenheit, a German physicist, proposed the Fahrenheit scale.

C. Kelvin Scale

Q1. What is the Kelvin scale?

Answer: The Kelvin scale is a temperature scale commonly used in scientific experiments.

Q2. What are the freezing and boiling points of water on the Kelvin scale?

Answer: Approximately 273.15 K and 373.15 K, respectively, under standard atmospheric pressure.

Q3. What is absolute zero?

Answer: Absolute zero is the lowest possible temperature, equal to 0 K.

Q4. What is the SI unit of temperature?

Answer: The kelvin (K) is the SI unit of temperature.

Q5. Who is associated with the Kelvin scale?

Answer: William Thomson, also known as Lord Kelvin, is associated with the Kelvin scale.

Comparison of Temperature Scales

Temperature	Celsius	Fahrenheit	Kelvin
Freezing point of water	0°C	32°F	273.15 K
Boiling point of water	100°C	212°F	373.15 K
Absolute zero	−273.15°C	−459.67°F	0 K

Note: These values are approximate where rounded. The textbook rounds absolute zero to about −273°C and −460°F.

4. Conversion of Temperature Scales (Pages 92–93)

We can convert temperature from one scale to another by using mathematical formulae.

Important Formulae

Celsius to Kelvin

\[K = {}^\circ C + 273.15 \]

Kelvin to Celsius

\[{}^\circ C = K - 273.15 \]

Fahrenheit to Celsius

$$ {}^\circ C = \frac{5}{9}({}^\circ F-32) $$

Celsius to Fahrenheit

$$ {}^\circ F = \frac{9}{5}({}^\circ C)+32 $$

Textbook Practice Questions (Page 93)

Q1. Convert 35°C into Kelvin.

Solution:

Formula:

$$ K = {}^\circ C + 273.15 $$
$$ K = 35 + 273.15 $$
$$ \boxed{308.15\ K} $$

Q2. Convert 370 K into Celsius.

Solution:

$$ {}^\circ C = K - 273.15 $$
$$ {}^\circ C = 370 - 273.15 $$
$$ \boxed{96.85^\circ C} $$

Q3. Convert 45°C into Fahrenheit.

Solution:

$$ {}^\circ F = \frac{9}{5}({}^\circ C)+32 $$
$$ {}^\circ F = \frac{9}{5}\times45+32 $$
$$ =81+32 $$
$$ \boxed{113^\circ F} $$

Q4. Convert 74°F into Celsius.

Solution:

$$ {}^\circ C = \frac{5}{9}({}^\circ F-32) $$
$$ {}^\circ C = \frac{5}{9}(74-32) $$
$$ =\frac{5}{9}\times42 $$
$$ \boxed{23.33^\circ C\text{ (approximately)}} $$

5. Thermometer (Page 94)

Q1. What is a thermometer?

Answer: A thermometer is a device used to measure temperature.

Q2. In which units can a thermometer measure temperature?

Answer: A thermometer may measure temperature in degrees Celsius, degrees Fahrenheit or kelvin, depending on its type and scale.

Q3. Name the two main types of thermometers described in the chapter.

Answer: The two main types are the clinical thermometer and the laboratory thermometer.

6. Clinical Thermometer (Pages 94–95)

Q1. What is a clinical thermometer?

Answer: A clinical thermometer is used to measure human body temperature.

Q2. What is the normal body temperature of a healthy human?

Answer: It is approximately 37°C or 98.6°F, although normal body temperature can vary from person to person.

Q3. What is the usual range of a clinical thermometer mentioned in the chapter?

Answer: The range is approximately 35°C to 42°C.

Q4. What is a mercury thermometer?

Answer: A mercury thermometer is a thermometer that uses mercury as its temperature-sensitive liquid.

Q5. Why are digital thermometers commonly used today?

Answer: Digital thermometers are easy to read and safer to handle because they do not contain mercury.

Q6. What is an infrared thermometer?

Answer: An infrared thermometer measures temperature without direct contact by detecting infrared radiation emitted by an object or body.

Q7. Can a clinical thermometer be used to measure the temperature of the atmosphere?

Answer: No. A clinical thermometer is designed for the limited range of human body temperature. A suitable air thermometer is used to measure atmospheric temperature.

How to Use a Digital Clinical Thermometer

Q8. Write the steps for measuring body temperature with a digital thermometer.

Answer:

1. Clean the thermometer according to its instructions.
2. Switch it on.
3. Wait until it indicates that it is ready.
4. Place it under the tongue or in the armpit, as instructed.
5. Wait for the beep or completed reading.
6. Read the temperature on the display.
7. Clean the thermometer after use.

Q9. Why should a thermometer be handled carefully?

Answer: It may break if dropped or struck against a hard surface. Broken glass or mercury can cause injury and contamination.

Activity: Measuring Family Members' Temperatures

Q10. Was the temperature of all family members the same throughout the week?

Answer: No, the readings may differ slightly between family members and from day to day.

Q11. Who had the highest temperature and on which day?

Answer: The answer depends on the temperature readings recorded during the activity.

Q12. Who had the lowest temperature and on which day?

Answer: The answer depends on the temperature readings recorded during the activity.

Q13. What may cause differences in body temperature?

Answer: Body temperature can vary because of age, physical activity, time of day, illness and other factors.

7. Laboratory Thermometer (Page 96)

Q1. What is a laboratory thermometer?

Answer: A laboratory thermometer is used to measure the temperature of liquids and other materials during scientific experiments.

Q2. What is the temperature range of the laboratory thermometer described in the chapter?

Answer: Its range is approximately −10°C to 110°C.

Q3. How does a laboratory thermometer work?

Answer: Its temperature-sensitive liquid rises or falls in a narrow tube as the temperature changes. We read the temperature against the scale.

Q4. What is the smallest division of the laboratory thermometer shown in the chapter?

Answer: The smallest division is 1°C.

Q5. How should we use a laboratory thermometer to measure the temperature of a liquid?

Answer:

1. Immerse the bulb completely in the liquid.
2. Do not allow the bulb to touch the bottom or sides of the container.
3. Keep the thermometer upright.
4. Wait for the reading to stabilise.
5. Read the scale at eye level while the thermometer is still in the liquid, if it is the type described in the textbook.

Q6. Why should the bulb not touch the sides or bottom of the beaker?

Answer: The container may be at a different temperature from the liquid, causing an inaccurate reading.

Q7. Why should we read a laboratory thermometer at eye level?

Answer: Reading at eye level helps prevent errors in reading the scale.

Q8. Can we use a laboratory thermometer to measure body temperature?

Answer: It is not recommended. A clinical thermometer is designed for measuring body temperature safely and appropriately.

8. Melting Point and Boiling Point of Water (Pages 96–97)

The chapter includes an activity to study the melting of ice and the boiling of water.

Q1. What is the melting point of ice under standard atmospheric pressure?

Answer: The melting point of ice is 0°C.

Q2. What is the boiling point of water under standard atmospheric pressure?

Answer: The boiling point of water is 100°C.

Q3. What happens when ice is heated?

Answer: Ice absorbs heat and changes from a solid into liquid water at its melting point.

Q4. What happens when water is heated to its boiling point?

Answer: Water boils and changes from a liquid into water vapour.

Q5. What is the aim of the activity on page 96?

Answer: The aim is to study the melting point of ice and the boiling point of water.

Q6. Name the apparatus required for this activity.

Answer: Ice cubes, a beaker, a thermometer, an iron stand, a tripod stand, a burner and a glass stirrer are required.

Q7. What observations should be recorded during the activity?

Answer:

- The temperature when the ice starts melting.
- The temperature when all the ice has melted.
- The temperature when the water starts boiling.
- The temperature when water starts changing into steam.

Q8. Why should the water be stirred while heating in the activity?

Answer: Stirring helps distribute heat more evenly throughout the water.

Q9. Does the temperature remain constant during the melting of pure ice at normal atmospheric pressure?

Answer: Yes. During melting, pure ice remains at about 0°C until it has melted completely.

Q10. Does the temperature remain constant during the boiling of pure water at normal atmospheric pressure?

Answer: Yes. During boiling, pure water remains at about 100°C while it changes into water vapour, under standard atmospheric pressure.

9. Air Temperature (Page 97)

Q1. What is air temperature?

Answer: Air temperature, also called atmospheric temperature, is the measure of how hot or cold the surrounding air is.

Q2. Which instrument is used to measure air temperature?

Answer: A suitable air thermometer is used to measure air temperature. Wall-mounted thermometers may show the approximate temperature of a room or its surroundings.

Q3. Why do we measure air temperature?

Answer: We measure air temperature to understand weather conditions and changes in the local climate.

Q4. How does air temperature usually change with the seasons?

Answer: Air temperature generally rises as summer approaches and falls during winter.

Q5. How can you check the air temperature in your area?

Answer: We can check a weather application, a weather report or a thermometer placed appropriately outdoors.

Q6. Who was Anna Mani?

Answer: Anna Mani (1918–2001) was an Indian physicist and meteorologist who contributed greatly to the development of weather-measuring instruments and the study of wind and solar energy.

HOTS: Important Thinking Questions

Q7. Why should we not judge the temperature of an object only by touching it?

Answer: Our sense of touch can be misleading because the sensation depends on the temperature of our hands and the previous contact with hot or cold objects.

Q8. Why is an infrared thermometer useful in public places?

Answer: It can measure temperature without direct contact, making screening quick and reducing the need to touch the person being checked.

Q9. Why is air temperature different at different times of the day?

Answer: The Sun heats the Earth's surface during the day, and the surface loses heat at night. Clouds, wind, humidity and other weather conditions also affect air temperature.

Q10. Why is it important to measure air temperature at a suitable place?

Answer: Direct sunlight or nearby hot surfaces can affect the reading. A thermometer should be placed in a suitable, shaded and ventilated location when measuring outdoor air temperature.

10. Additional Important Questions and Answers

A. Temperature and Its Importance

Q1. What is the difference between heat and temperature?

Answer: Temperature tells us how hot or cold a body is. Heat is energy transferred from a hotter body to a colder body because of a temperature difference.

Q2. Which body has a higher temperature: a cup of hot tea or a glass of cold water?

Answer: The cup of hot tea has a higher temperature.

Q3. Why is temperature important in daily life?

Answer: Temperature helps us check body health, cook food, conduct scientific experiments and understand weather conditions.

Q4. What is the freezing point of water in Celsius?

Answer: 0°C.

Q5. What is the boiling point of water in Celsius?

Answer: 100°C under standard atmospheric pressure.

B. Temperature Scales

Q1. Which temperature scale is commonly used in India for everyday temperatures?

Answer: The Celsius scale.

Q2. Which temperature scale is used in many scientific calculations?

Answer: The Kelvin scale.

Q3. Why is the Kelvin scale important in science?

Answer: It is the SI temperature scale and begins at absolute zero.

Q4. What is the relation between Celsius and Kelvin?

Answer: Kelvin temperature equals Celsius temperature plus 273.15.

Q5. What is the relation between Celsius and Fahrenheit?

Answer: Fahrenheit temperature equals nine-fifths of the Celsius temperature plus 32.

C. Thermometers

Q1. Write two differences between a clinical thermometer and a laboratory thermometer.

Answer:

Clinical thermometer	Laboratory thermometer
Used to measure human body temperature	Used to measure temperatures during experiments
Has a limited range suitable for body temperature	Usually has a wider range for laboratory work

Q2. Why are mercury thermometers less commonly used today?

Answer: Mercury is toxic, and broken mercury thermometers can cause health and environmental hazards. Digital thermometers are generally safer.

Q3. Why must we follow the instructions supplied with a digital thermometer?

Answer: Different thermometers may require different methods of use. Following the instructions helps ensure a safe and reliable reading.

Q4. Why should we not hold a thermometer by its bulb while taking a reading?

Answer: Touching the bulb can affect the reading, and careless handling may damage the thermometer.

Q5. What is the difference between contact and non-contact thermometers?

Answer: A contact thermometer measures temperature through contact with the object or body. A non-contact thermometer, such as an infrared thermometer, measures temperature from a distance.

D. Numerical Questions for Practice

Q1. Convert 25°C into Kelvin.

Solution:

$$K=25+273.15$$
$$\boxed{298.15\ K}$$

Q2. Convert 100°C into Fahrenheit.

Solution:

$$^\circ F=\frac95(100)+32$$
$$=180+32$$
$$\boxed{212^\circ F}$$

Q3. Convert 32°F into Celsius.

Solution:

$$^\circ C=\frac59(32-32)$$
$$\boxed{0^\circ C}$$

Q4. Convert 300 K into Celsius.

Solution:

$$^\circ C=300-273.15$$
$$\boxed{26.85^\circ C}$$

Q5. Convert 20°C into Fahrenheit.

Solution:

$$^\circ F=\frac95(20)+32$$
$$=36+32$$
$$\boxed{68^\circ F}$$

11. Multiple-Choice Questions (MCQs)

Chapter Practice Test

0/12 answered

1. What is temperature?

A. The mass of a body
B. The degree of hotness or coldness
C. The volume of a body
D. The length of a body

2. What is the SI unit of temperature?

A. Celsius
B. Fahrenheit
C. Kelvin
D. Degree

3. At what temperature does water freeze on the Celsius scale?

A. 0°C
B. 32°C
C. 100°C
D. 273°C

4. What is the normal human body temperature approximately?

A. 0°C
B. 25°C
C. 37°C
D. 100°C

5. Which instrument is used to measure human body temperature?

A. Laboratory balance
B. Clinical thermometer
C. Measuring tape
D. Barometer

6. Which thermometer measures temperature without direct contact?

A. Mercury clinical thermometer
B. Laboratory thermometer
C. Infrared thermometer
D. Room thermometer only

7. Which temperature is equal to 100°C?

A. 32°F
B. 98.6°F
C. 212°F
D. 273.15°F

8. What is the freezing point of water in Fahrenheit?

A. 0°F
B. 32°F
C. 100°F
D. 212°F

9. What is the boiling point of water in Kelvin?

A. 273.15 K
B. 300 K
C. 373.15 K
D. 100 K

10. Who is associated with the Celsius scale?

A. Anders Celsius
B. Anna Mani
C. Lord Kelvin only
D. Isaac Newton

11. Which thermometer is used for scientific experiments?

A. Clinical thermometer
B. Laboratory thermometer
C. Only an infrared thermometer
D. None of these

12. What is the approximate melting point of ice at standard pressure?

A. −10°C
B. 0°C
C. 50°C
D. 100°C
Check Answers

12. Fill in the Blanks

1. The measure of the hotness or coldness of a body is called temperature.
2. The SI unit of temperature is kelvin.
3. The freezing point of water on the Celsius scale is 0°C.
4. The boiling point of water on the Celsius scale is 100°C.
5. A device used to measure temperature is called a thermometer.
6. A clinical thermometer is used to measure body temperature.
7. A laboratory thermometer is used during scientific experiments.
8. The approximate normal human body temperature is 37°C.
9. The lowest possible temperature is called absolute zero.
10. The instrument used to measure atmospheric temperature is a suitable air thermometer.

13. True or False

1. Our sense of touch always measures temperature accurately. — False
2. The SI unit of temperature is kelvin. — True
3. Water freezes at 0°C under standard atmospheric pressure. — True
4. Water boils at 100°C under standard atmospheric pressure. — True
5. A clinical thermometer is designed to measure the temperature of a laboratory furnace. — False
6. Digital thermometers are generally safer to handle than mercury thermometers. — True
7. A laboratory thermometer's bulb should touch the bottom of the beaker while measuring a liquid. — False
8. An infrared thermometer can measure temperature without direct contact. — True
9. Air temperature changes with weather conditions and seasons. — True
10. Anna Mani contributed to the development of weather-measuring instruments in India. — True

14. Quick Revision

- Temperature: Measure of the degree of hotness or coldness of a body.
- Thermometer: An instrument used to measure temperature.
- Celsius scale: Water freezes at 0°C and boils at 100°C under standard atmospheric pressure.
- Fahrenheit scale: Water freezes at 32°F and boils at 212°F.
- Kelvin scale: The SI temperature scale.
- Clinical thermometer: Measures human body temperature.
- Laboratory thermometer: Measures temperature during experiments.
- Infrared thermometer: Measures temperature without direct contact.
- Melting point of ice: Approximately 0°C at standard pressure.
- Boiling point of water: Approximately 100°C at standard pressure.
- Air temperature: Temperature of the surrounding air.
- Anna Mani: Indian scientist known for her contributions to meteorology and weather instruments.

Exam preparation tip: First learn the definitions and differences between the three temperature scales and the two main types of thermometers. Then practise all four conversion formulae and the textbook numerical questions.